

Question 7

Q: A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Theoretical Concept: Levels of Computer Vision

Quantization is the process of mapping continuous intensity values (voltages from the sensor) into a discrete set of digital values (e.g., 0 to 255). It determines the **color depth** or intensity resolution of the image.

Impact: If quantization levels are too low (e.g., 2-bit or 4-bit), smooth gradients become blocky, creating "false contours" or banding artifacts. High quantization preserves fine texture details.

OpenCV Python Solution

```
# =====
# AIM:
# Demonstrate Quantization in Digital Image Processing
# =====

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload grayscale image
print("Upload a grayscale image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

# Function for quantization
def quantize(image, levels):
```

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        step = 256 // levels
        return (image // step) * step

# Different quantization levels
q256 = img
q64 = quantize(img, 64)
q16 = quantize(img, 16)
q4 = quantize(img, 4)

# Display images
plt.figure(figsize=(12,8))

titles = ["Original (256 Levels)",
          "64 Levels",
          "16 Levels",
          "4 Levels"]

images = [q256, q64, q16, q4]

for i in range(4):
    plt.subplot(2,2,i+1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis("off")

plt.show()

# Output
print("\n===== QUANTIZATION REPORT =====")
print("Original Image      : 256 Gray Levels")
print("Image 1              : 64 Gray Levels")
print("Image 2              : 16 Gray Levels")
print("Image 3              : 4 Gray Levels")
print("-----")
print("Observation:")
print("- Higher gray levels preserve image detail.")
print("- Lower gray levels reduce detail.")
print("- Very low levels create visible intensity bands.")
print("=====")

```

Expected Output

The plot compares a perfectly smooth glowing sphere with a heavily distorted sphere made of concentric rigid rings. This clearly visualizes the physical impact of low quantization limits.